0826 labmeeting

윤현석

August 25, 2025

1 previous

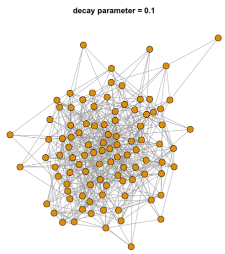
simulation data 에 대한 LSM, LSJM 의 rmse 구하기 계속 진행중

2 simulation setting

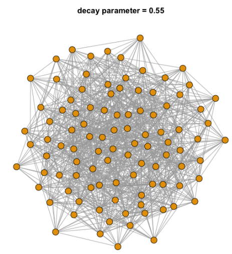
```r
sample_with_gwesp <- function(gwesp_coef, theta_edges = 1, decay = 0.5) {
  sims <- simulate(
    g0 ~ edges + gwesp(decay, fixed = TRUE),
    coef = c(theta_edges, gwesp_coef),
    nsim = 1,
    output = "network",
    control = control.simulate(
      MCMC.burnin = 1e5,
      MCMC.interval = 1e3
    )
  )
  return(sims)
}

# only control decay parameter: 0.1, 0.15, ... , 1.2
decay_values <- seq(from = 0.1, to = 1.2, by = 0.05)
n <- length(decay_values)
nets <- list()
for(i in 1:n){
  nets[[i]] <- sample_with_gwesp(gwesp_coef = 1.0, theta_edges = -3.5, decay =
    decay_values[i])
}
```

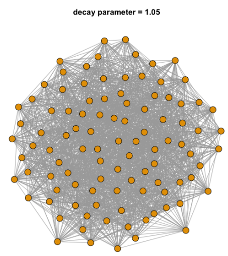
2.1 plot



(a) Network with decay param: 0.1



(b) Network with decay param: 0.55

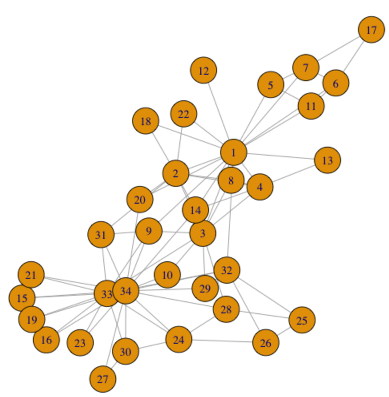


(c) Network with decay param: 1.05

Figure 1: 세 개의 네트워크 비교

```
decay_par =  0.10 | edges = 554 | transitivity = 0.166 | triangles = 1041
decay_par =  0.55 | edges = 1016 | transitivity = 0.225 | triangles = 4578
decay_par =  1.05 | edges = 1885 | transitivity = 0.379 | triangles = 26589
```

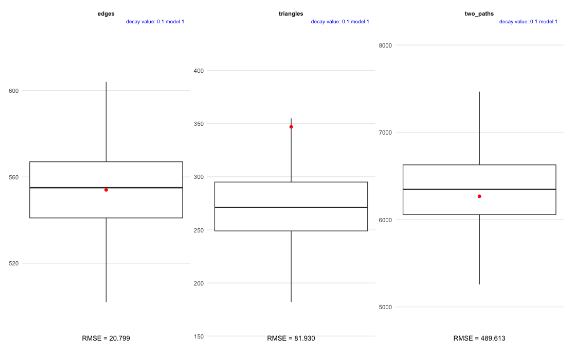
2.2 real data - karate club



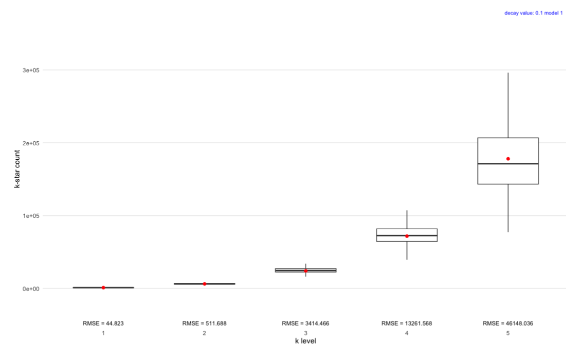
3 result

3.1 LSM (without communicability distance (standard scaling))

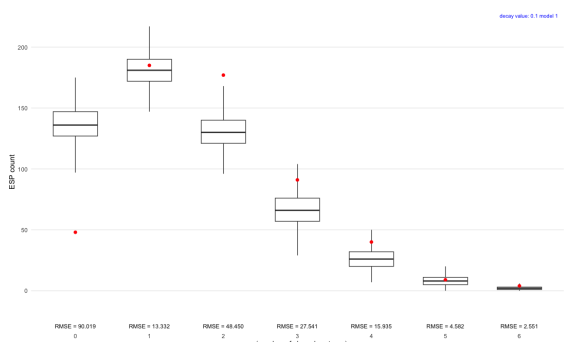
3.1.1 network with decay param: 0.1



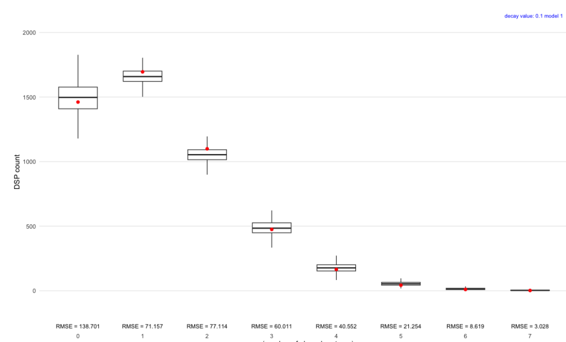
(a) edge, triangle, 2 path count



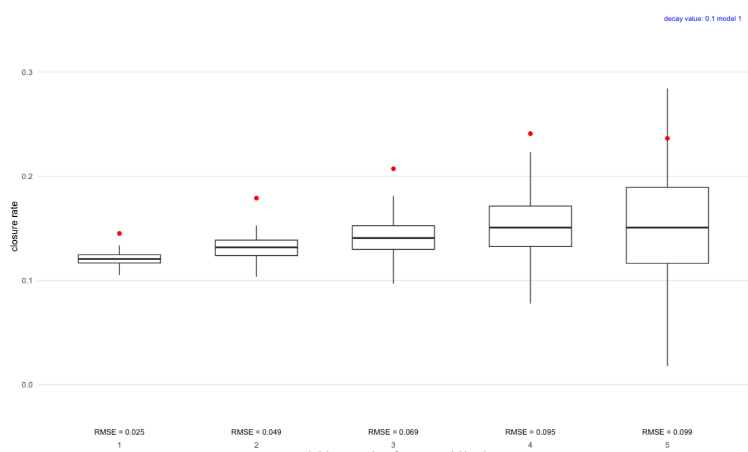
(b) k-star count (1:5)



(c) dsp count



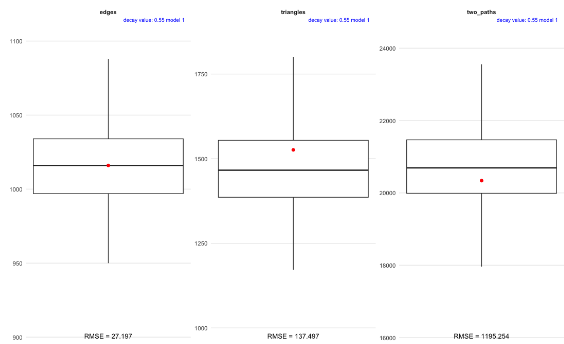
(d) esp count



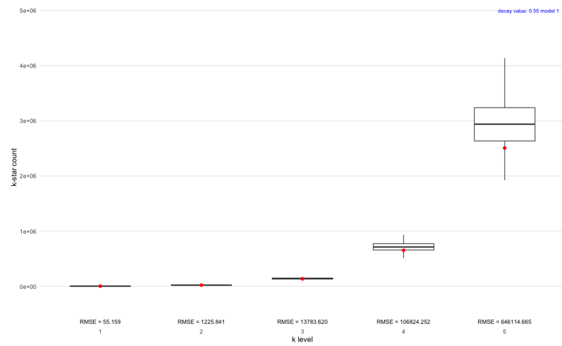
(e) esp/dsp rate

Figure 2: motif 별 boxplot, rmse 정리

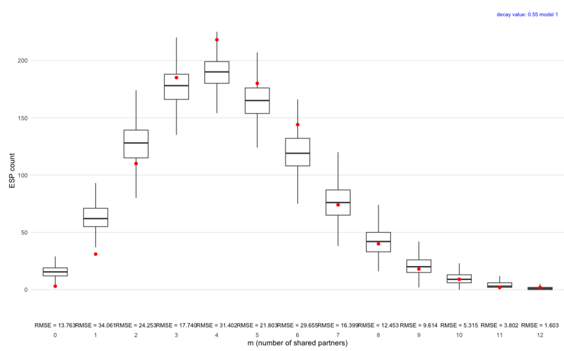
3.1.2 network with decay param: 0.55



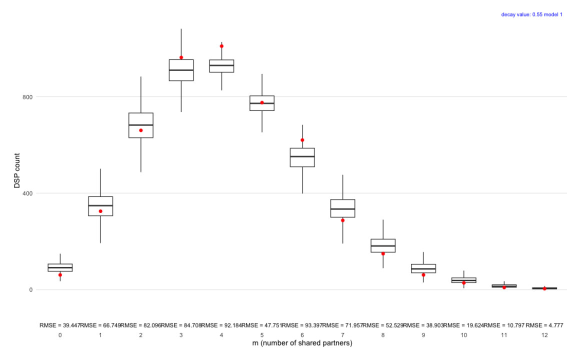
(a) edge, triangle, 2 path count



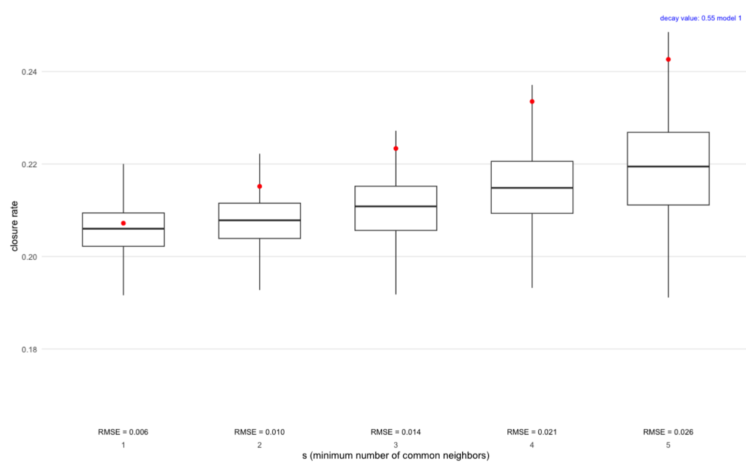
(b) k-star count (1:5)



(c) dsp count



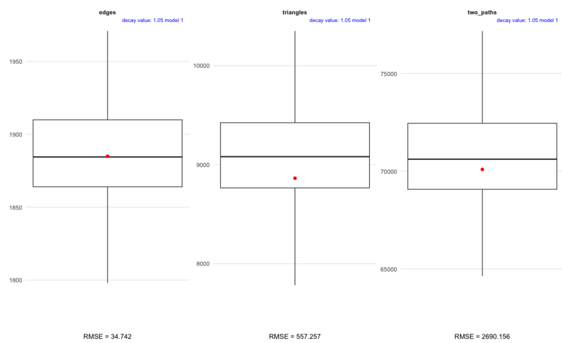
(d) esp count



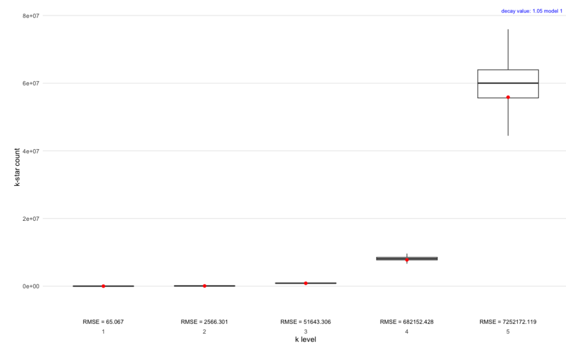
(e) esp/dsp rate

Figure 3: motif 별 boxplot, rmse 정리

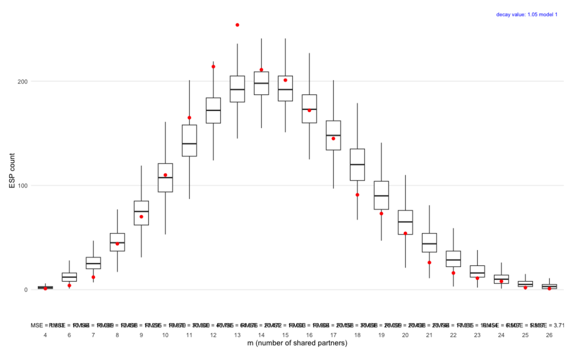
3.1.3 network with decay param: 1.05



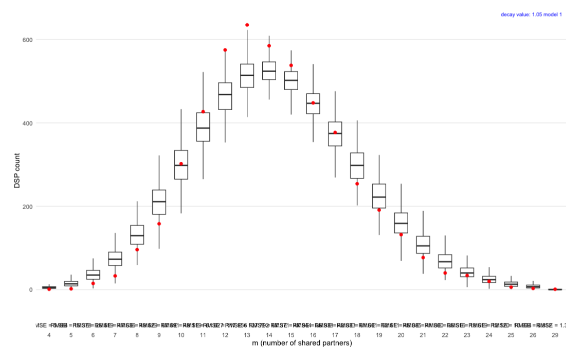
(a) edge, triangle, 2 path count



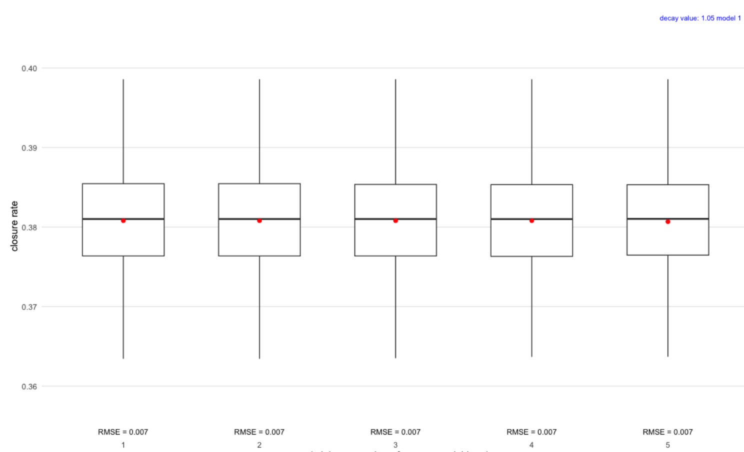
(b) k-star count (1:5)



(c) dsp count



(d) esp count

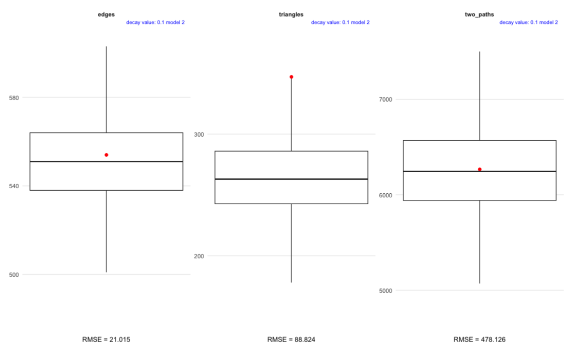


(e) esp/dsp rate

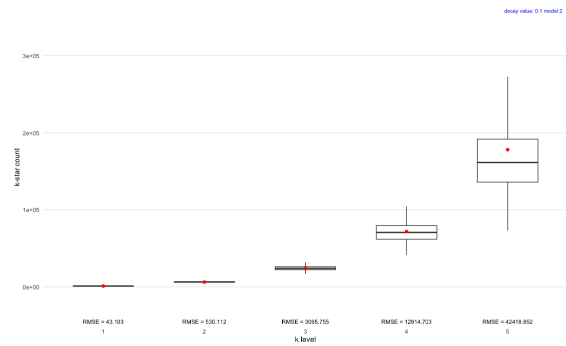
Figure 4: motif 별 boxplot, rmse 정리

3.2 LSM (with communicability distance (standard scaling))

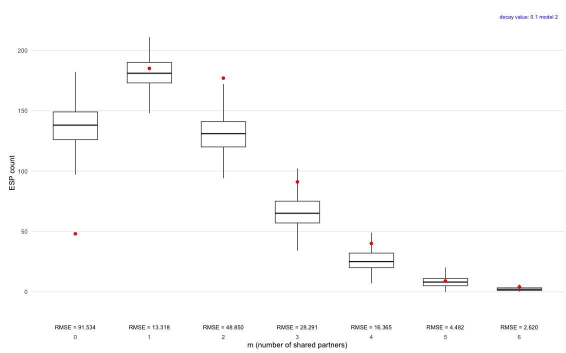
3.2.1 network with decay param: 0.1



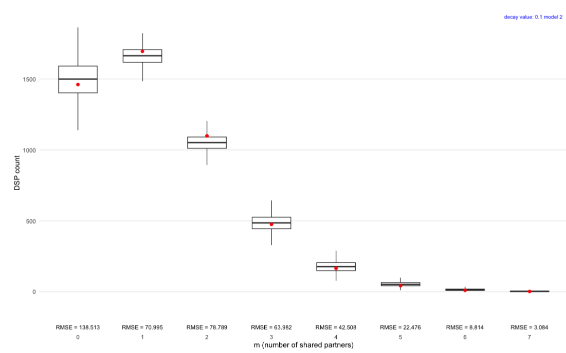
(a) edge, triangle, 2 path count



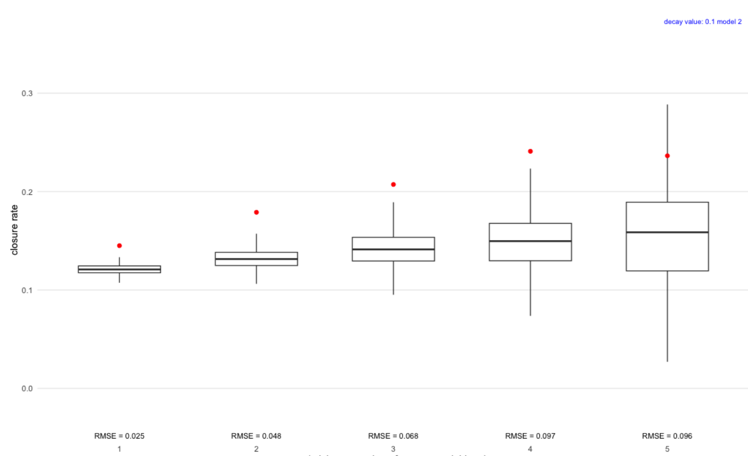
(b) k-star count (1:5)



(c) dsp count



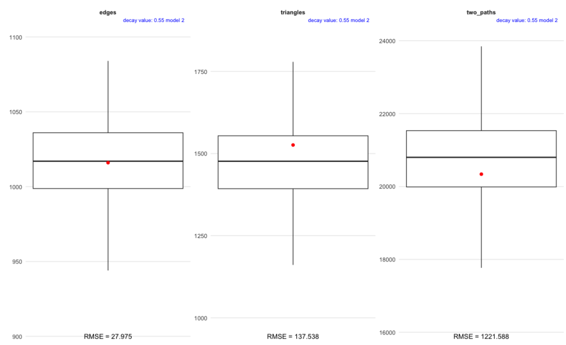
(d) esp count



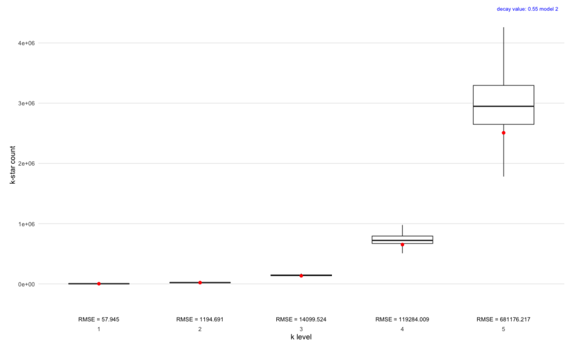
(e) esp/dsp rate

Figure 5: motif 별 boxplot, rmse 정리

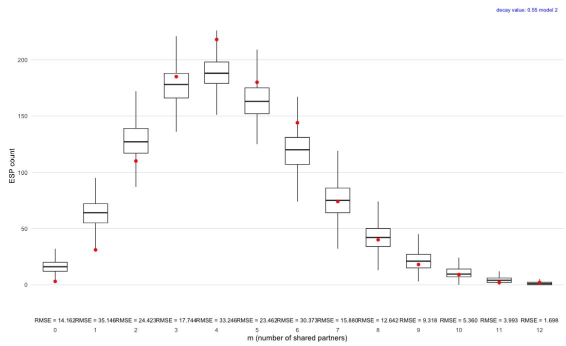
3.2.2 network with decay param: 0.55



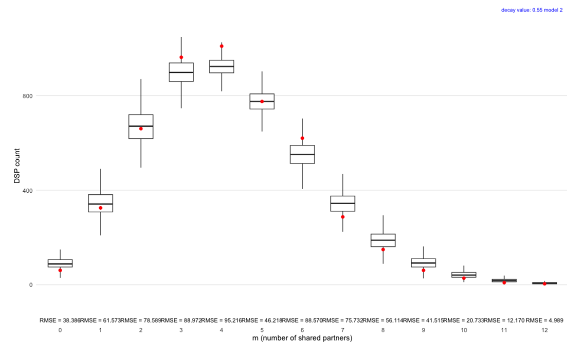
(a) edge, triangle, 2 path count



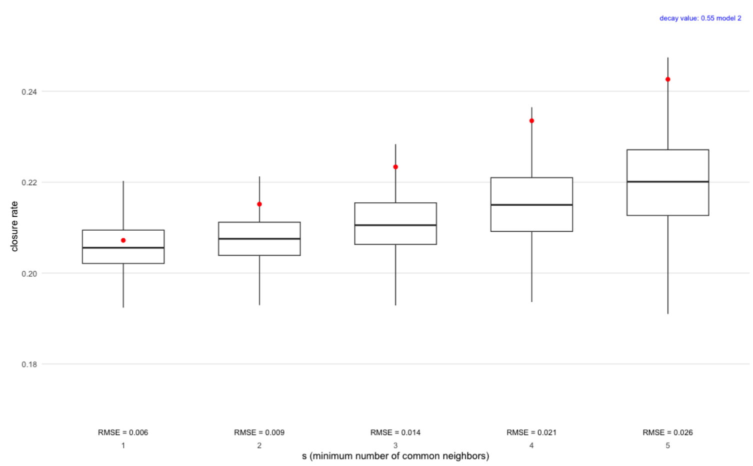
(b) k-star count (1:5)



(c) dsp count



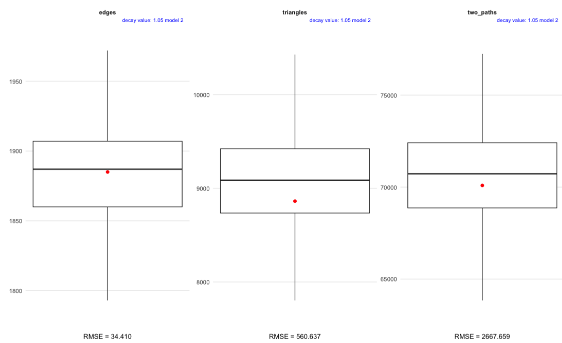
(d) esp count



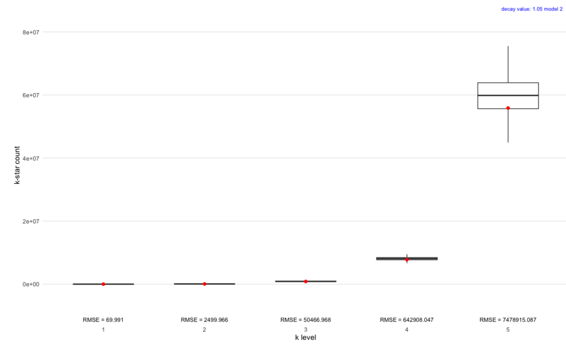
(e) esp/dsp rate

Figure 6: motif 별 boxplot, rmse 정리

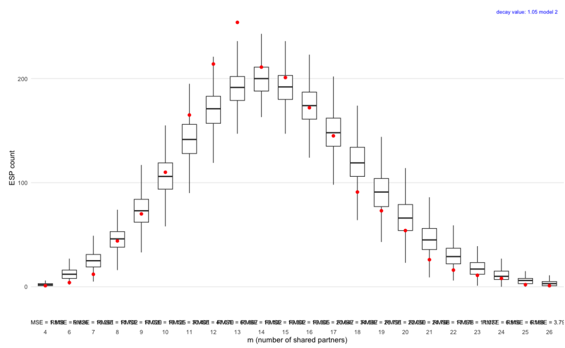
3.2.3 network with decay param: 1.05



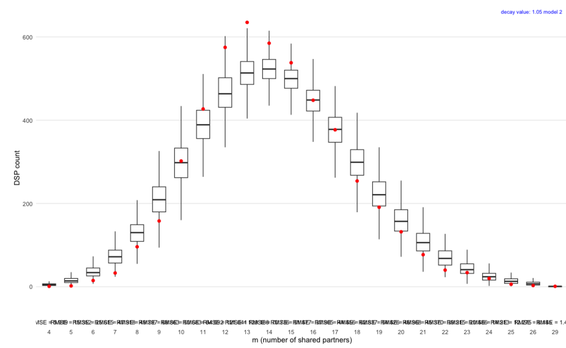
(a) edge, triangle, 2 path count



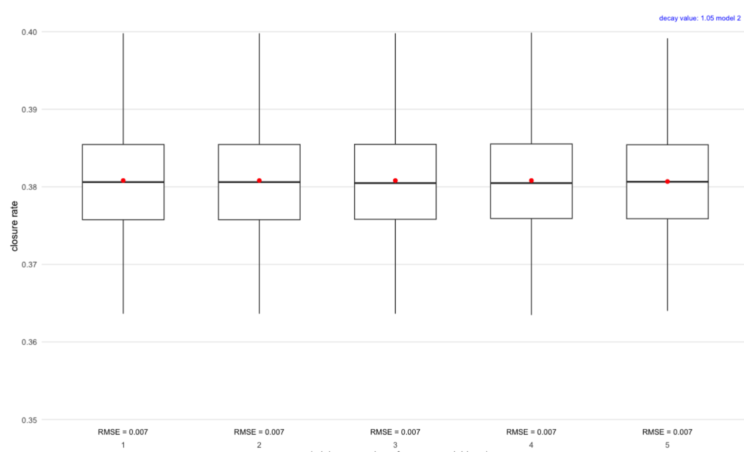
(b) k-star count (1:5)



(c) dsp count



(d) esp count

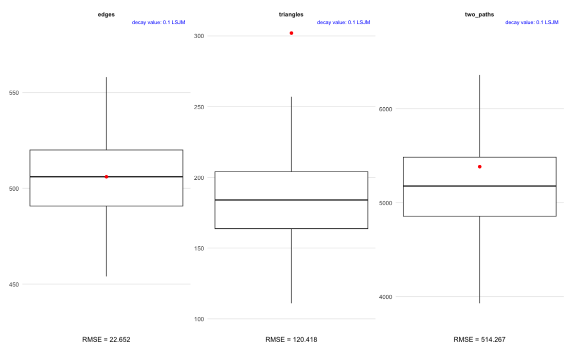


(e) esp/dsp rate

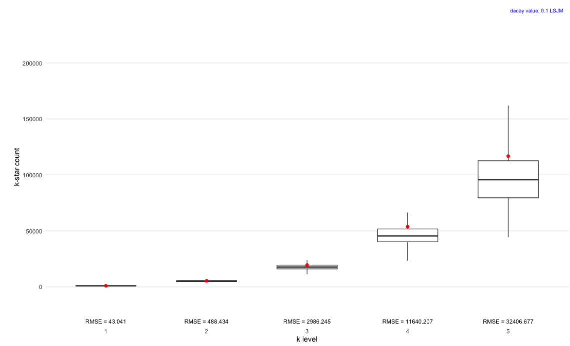
Figure 7: motif 별 boxplot, rmse 정리

3.3 LSJM with communicability network

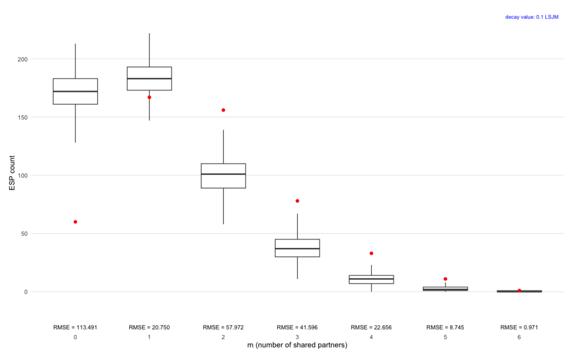
3.3.1 network with decay param: 0.1



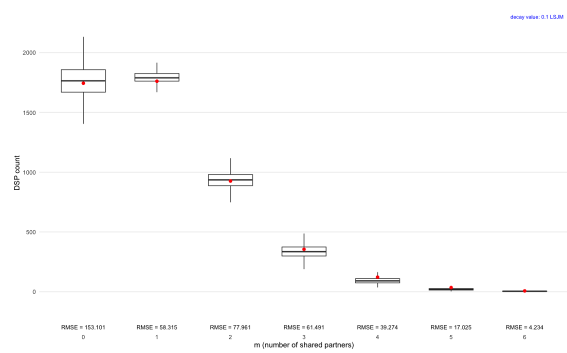
(a) edge, triangle, 2 path count



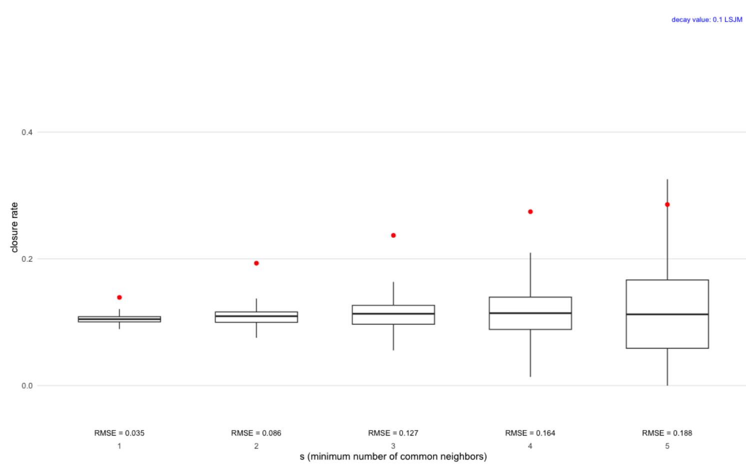
(b) k-star count (1:5)



(c) dsp count



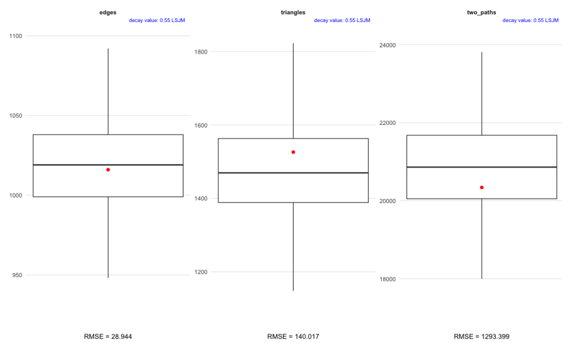
(d) esp count



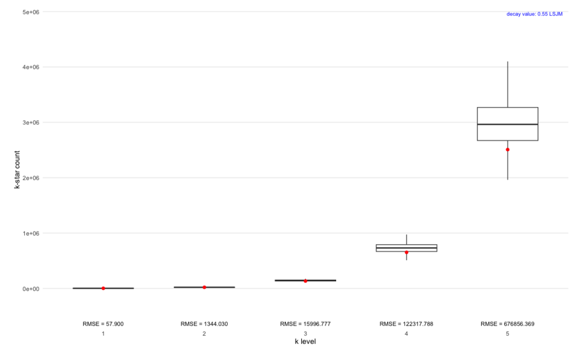
(e) esp/dsp rate

Figure 8: motif 별 boxplot, rmse 정리

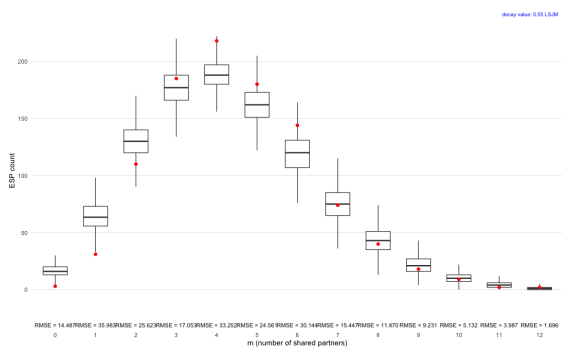
3.3.2 network with decay param: 0.55



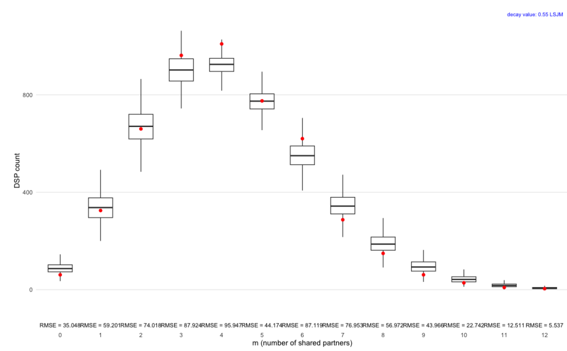
(a) edge, triangle, 2 path count



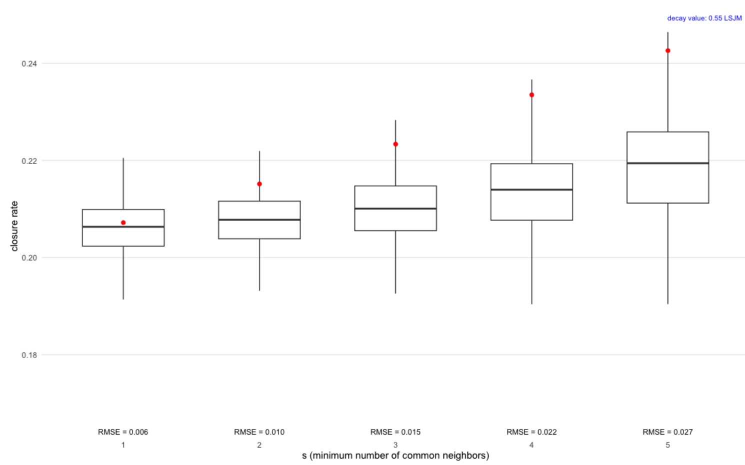
(b) k-star count (1:5)



(c) dsp count



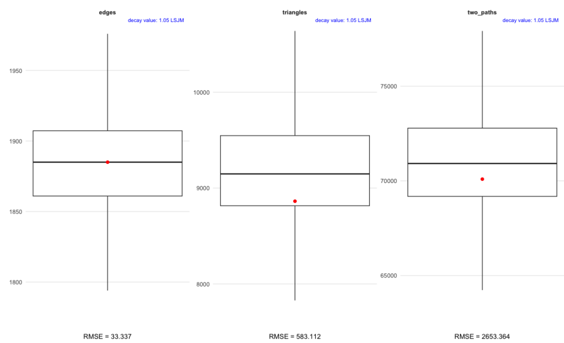
(d) esp count



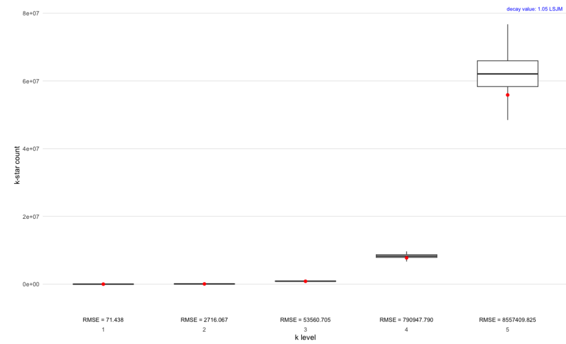
(e) esp/dsp rate

Figure 9: motif 별 boxplot, rmse 정리

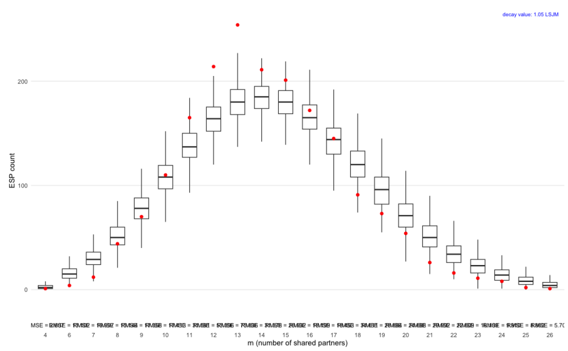
3.3.3 network with decay param: 1.05



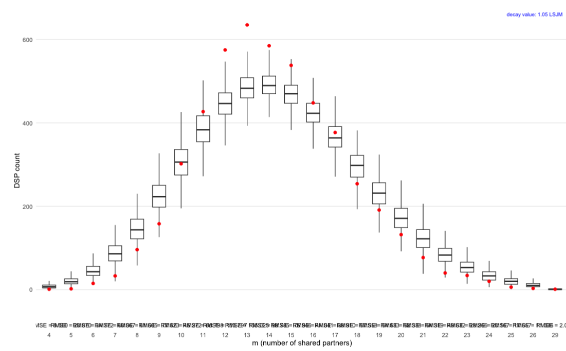
(a) edge, triangle, 2 path count



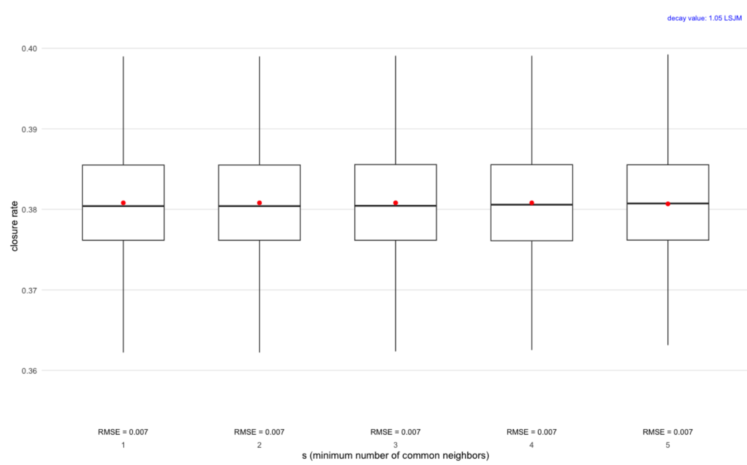
(b) k-star count (1:5)



(c) dsp count



(d) esp count



(e) esp/dsp rate

Figure 10: motif 별 boxplot, rmse 정리

결론적으로, 비슷한 성능을 보입니다.

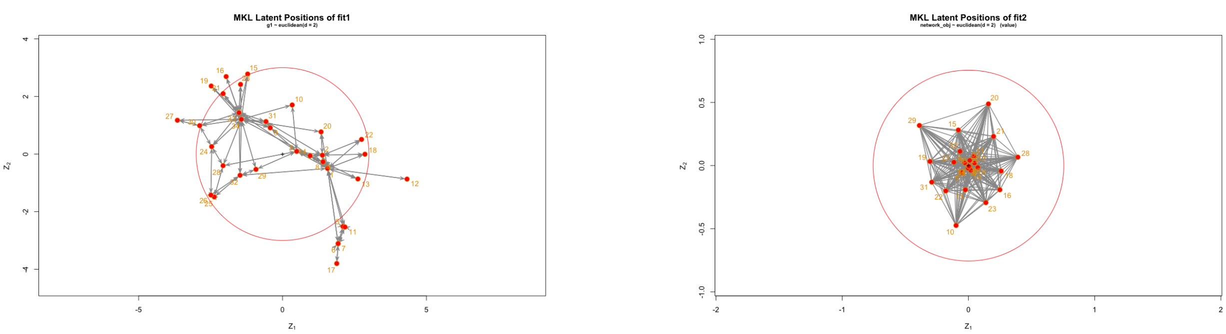
4 communicability (distance)

그렇다면 왜 비슷한 성능을 보일까?

- 1. communicability 가 원래 network 와 거의 같은 구조를 보이기 때문에 정보의 추가가 없다.
- 2. communicability 의 정보가 의미 없는 형태를 가진다.

이것을 확인해 보기 위해서 communicability 만을 사용하여 LSM 을 돌린 결과와 adjacency matrix 만을 사용하여 돌린 결과를 비교해 보았습니다. 빠른 확인을 위해 karate network 로 수행하였습니다.

4.1 example of karate club data



(a) LSM with adjacency matrix

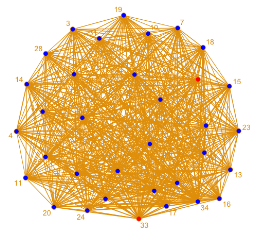
(b) LSM with communicability

Figure 11: LSM result for karate network

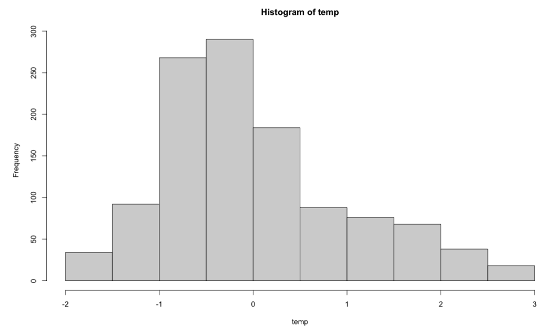
communicability 를 사용한 LSM 결과는 원점으로 대부분의 node 가 모여 있는 구조로 보였습니다. karate network 는 두 개의 cluster 가 있으므로 communicability 를 사용해도 두 개의 cluster 구조가 어느 정도 복원되어야 하는 것이 아닌가? 라는 의문이 있음.

communicability distance 로 표현한 karate network 의 모습입니다.

4.2 example of karate club data



(a) network visualization of karate communicability network



(b) degree distribution

Figure 12: LSM result for karate network

standard scaling 된 communicability distance 가 충분히 각 node 간의 차이를 반영하지 못 하고 있는 것은 아닌가?

그런데, 임의의 값으로 threshold 를 설정하고 계산하면 network 구조가 어느 정도는 보입니다. 지금 생각하고 있는 것은 communicability 가 network 상에서 떨어져 있는 node 들에 대해서 지나치게 높은 값을 주고 있는 것은 아닌가? 하는 생각입니다. 그래서 LSM 결과가 그냥 원점으로 모이게 되고, 그래서 communicabilty effect 가 마치 상수항처럼 작동하여, LSJM 의 성능에 아무런 기여를 하고 있지 못하는 게 아닌가?

References